# Preliminary Design Tool for Flying Wing Unmanned Aerial Systems

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Thesis contribution: an integrated preliminary-design workflow that keeps geometry, aerodynamics, structure, mission, and export tied to one aircraft definition.

## Abstract

The Flying Wing Tool is a preliminary-design environment for tailless aircraft built around one shared project model. The research problem addressed here is workflow fragmentation: geometry, aerodynamics, structure, mission, and export are often separated across different tools, which slows iteration and allows assumptions to drift. The implemented environment keeps those tasks tied to one aircraft definition. A representative blended-wing case study plus an independent BARAM CFD path were used to show both the value and the present limits of that workflow.

## Research Gap and Contribution

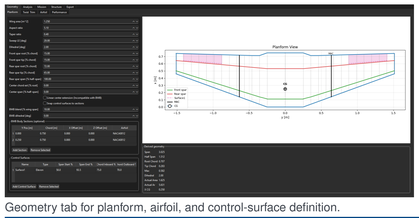
- Flying-wing design is tightly coupled: planform and twist changes affect trim, loading, structure, and manufacturability together.
- Fragmented tool chains slow iteration because geometry and assumptions are repeatedly re-entered.
- The main contribution is software integration: one serialized project state supports geometry, aerodynamics, mission, structure, and export.

## Implemented Workflow

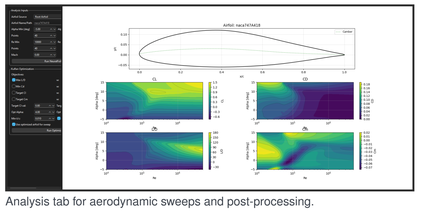
- GUI tabs: Geometry, Analysis, Mission, Structure, and Export.
- Shared state: one project object stores wing definition, trim assumptions, mission settings, and outputs.
- Service layer: geometry/aerodynamics, mission/propulsion, wingbox-oriented structure, and export.
- Reproducibility path: `run_full_analysis.py` reloads saved projects for repeatable reruns.

### Capability snapshot

| Module | State | Role in the current thesis scope |
|---|---|---|
| **Geometry** | Active | Parametric planform, airfoil, control-surface, and section generation |
| **Analysis** | Active | Aero polars, twist synthesis, and performance metrics |
| **Mission** | Active | Stored mission simulation and phase-level summaries |
| **Structure** | Active | Wingbox sizing, margins, deflection, and feasibility checks |
| **Export** | Active | CAD/manufacturing-oriented geometry handoff |
| **Batch rerun** | Active | File-based reproducibility from saved projects |




Geometry tab for planform, airfoil, and control-surface definition.




Analysis tab for aerodynamic sweeps and post-processing.

## Why the Integrated Loop Matters

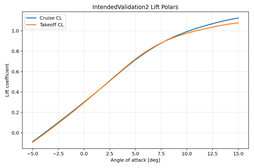
- The same spanwise sections drive aerodynamic loading, structural sizing, and exported geometry.
- Twist optimization is written back into the same project state used for later structural and mission analysis.
- One saved case can expose aerodynamic promise together with a governing structural or drag limitation.

## Representative Case: IntendedValidation2

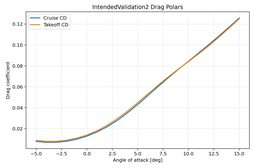
| | | | |
|---|---|---|---|
| **Planform area** | 1.625 m$^2$ | **Cruise speed** | 19.95 m/s |
| **Span** | 3.025 m | **Takeoff speed** | 13.34 m/s |
| **Aspect ratio** | 5.63 | **Cruise angle of attack** | 0.00 deg |
| **Gross weight** | 12.0 kg | **Takeoff angle of attack** | 5.09 deg |
| **Lift target** | Bell-shaped | **Control surface** | Single elevon set |
| **Static margin** | 8% | **Centerbody type** | Blended body sections |

Representative design inputs and operating-point conditions for the `IntendedValidation2` case used throughout the poster.

## Aerodynamic and Twist Results




Lift polar.




Drag polar.

- Cruise operating point: $C_L = 0.300$, $C_D = 0.0129$, $L/D = 23.21$ at $0^\circ$ angle of attack.
- Takeoff operating point: $C_L = 0.730$, $C_D = 0.0456$, $L/D = 16.02$ at $5.09^\circ$ angle of attack.
- Optimized twist still trends from approximately $+3.29^\circ$ inboard to $-1.48^\circ$ outboard, but the figure is omitted here to prioritize space for validation and case-definition content.

## Integrated Structure and Mission Outputs

### Structure

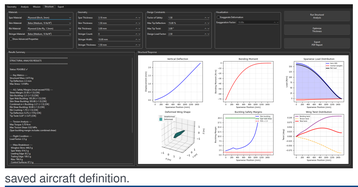
| | |
|---|---|
| **Structural mass** | 2.363 kg |
| **Tip deflection** | 2.27 mm |
| **Tip twist** | 0.197 deg |
| **Min rib crushing margin** | 0.97 |
| **Feasibility flag** | False |

The structure is globally light and stiff, but a local rib-crushing mode remains slightly below unity and governs feasibility.

### Mission

| | |
|---|---|
| **Mission completion** | 9 of 9 phases |
| **Distance** | 1283 m |
| **Energy used** | 6.64 Wh |
| **Final SOC** | 78.1% |
| **Max electrical power** | 2.30 kW |

The same aircraft definition can be evaluated from an energy-use and mission-feasibility perspective without leaving the project framework.

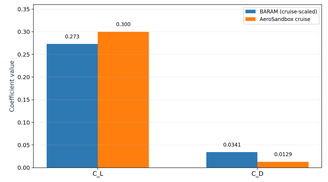


Structural-analysis interface used to inspect stress, margins, and feasibility for the same saved aircraft definition.

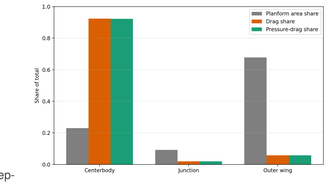
## Case Study Interpretation

The case study shows why the shared-state workflow matters. The same saved aircraft produces a coherent cruise solution, a usable twist distribution, a lightweight but not yet fully feasible structure, and a mission-level energy summary. The tool therefore surfaces coupled design consequences rather than isolated outputs.

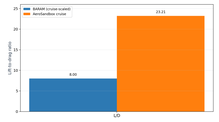
## Validation: BARAM CFD Cross-Check




Separated coefficient comparison for $C_L$ and $C_D$.




Projected planform share versus split-patch drag share.




Separated $L/D$ comparison so the coefficient bars are not visually compressed.

- Independent CFD path: incompressible RANS, standard $k$-$\epsilon$, symmetry half-model, convergence after 604 SIMPLE iterations.
- After rescaling to the thesis reference quantities, lift is directionally close: $C_L = 0.273$ in BARAM versus $0.300$ in the repository cruise analysis.
- Drag differs materially: $C_D = 0.0341$ in BARAM versus $0.0129$ in the repository cruise analysis, reducing $L/D$ from 23.21 to 8.00.
- The CFD drag is pressure dominated: 77.2% pressure drag and 22.8% viscous drag.
- The centerbody plus junction occupy only 32.2% of projected planform area but account for 94.3% of the drag share, indicating a centerbody-dominated pressure-drag mechanism.

## Mesh-Sensitivity Study

| Condition | Cells (coarse → medium → medium 2 → fine) | $\Delta C_D$ | $\Delta C_L$ | Read |
|---|---|---|---|---|
| **Cruise** | 0.423M → 2.159M → 2.773M → 8.505M | 2.02% | 0.57% | Mixed |
| **Takeoff** | 0.423M → 2.159M → 2.773M → 33.901M | 3.13% | 0.58% | Sensitive |

- Former "fine" meshes are relabeled Medium 2 because they sat very close to the medium meshes in total cell count; former "extra-fine" meshes are reported as Fine.
- The reported $\Delta C_D$ and $\Delta C_L$ values are the final Medium 2 → Fine changes, so this section is presented as mesh sensitivity rather than formal grid independence.
- Retained wall-layer cells remained zero, so the study captures coefficient sensitivity of the present setup rather than full boundary-layer independence.

## Claim Boundary and Next Steps

- The main result is an integrated, traceable preliminary-design environment for flying-wing aircraft rather than a stand-alone new solver.
- The validation evidence supports a narrow claim: the present low-order workflow preserves lift and trim trends better than absolute drag for thick blended-centerbody configurations.
- The dominant missing term is centerbody-dominated pressure and interference drag rather than outer-wing skin friction alone.
- Next work should focus on stronger centerbody drag modeling, retained wall-layer verification, and comparison against additional published or experimental cases.

## Selected References

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